The Impact of Active Algorithm Control on Social Media Satisfaction: An Experimental Study on Active Social Media Users in CUHK-Shenzhen

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Abstract

This study examines the effects of active algorithm control on content personalization and user satisfaction among active social media users. This research recruited 50 undergraduate students from CUHKSZ to conduct a one-week randomized trial. This research adopts a mixed-method approach. Quantitative data are collected from a pre-test and a post-test. Qualitative data are collected from optional open-ended questions. The result shows that active algorithm control significantly changes the content recommendations. Furthermore, increased control was associated with higher satisfaction, mainly through enhanced perceived autonomy, improved relevance of content, and reduced exposure to unwanted information. The study contributes to the literature on user agency in algorithmic systems and offers practical insights for the design of more transparent and empowering social media interfaces.

Introduction

In the contemporary digital age, more and more social media platforms are using algorithms to curate and prioritize content. Although these algorithms are designed to maximize user engagement and platform stickiness, they often lead to passive consumption and may create phenomena such as “filter bubbles” and “echo chambers”. The current studies highlight the effects of algorithms on users, pointing out issues such as biased content distribution, polarization, and the shaping of public opinion. In contrast, less attention is paid to how users can actively shape algorithms. While features like “Not Interested” or “Dislike” are commonly available on major platforms, empirical understanding of their efficacy in altering content outcomes and their psychological impact on users remains limited. This research aims to fill that gap by addressing the following questions.

Research Questions

R.Q.1. To what extent does the algorithm control lead to content changes?

R.Q.2. In which perspective does increased algorithm control lead to higher satisfaction?

Literature Review

The scholarly discourse around algorithms and social media users can mainly be divided into two parts. One is the impact of algorithmic systems on users' behavior and cognition. Another is users' perceptions and behaviors toward these systems. A large body of research has demonstrated the powerful influence of algorithms on user experience. Algorithms are instrumental in creating what Pariser (2011) termed the "filter bubble," a personalized informational universe that limits exposure to divergent viewpoints. This is often driven by engagement-based metrics. These metrics prioritize content aligning with a user's established preferences, thereby potentially amplifying cognition bias and societal polarization (Sirbu et al., 2019). Research on platforms like Twitter and Facebook demonstrates that algorithms can significantly influence public perceptions (Dujeancourt & Garz, 2023).

Meanwhile, other research has begun to investigate users' awareness of and interactions with these systems. The concept of "Algorithmic Experience" (AX) emphasizes the subjective, perceptual dimension of users' encounters with algorithmic curation. Studies find that many users have only a limited understanding of how their information feeds are organized. And even when they are aware of this, the users tend to believe that the possibility of meaningful intervention is limited (Alvarado & Waern, 2018). This often leads to feelings of frustration and powerlessness, especially when users encounter irrelevant or undesirable content (Yang et al., 2021). Some researchers argue that platform interfaces often obscure control mechanisms, placing them in sub-menus or providing minimal feedback, which reinforces a default state of passive consumption (Bernstein et al., 2023).

Despite extensive research on users and algorithms, a gap still exists. While we know much about algorithmic effects and user frustration, the empirical research on the outcomes of explicitly encouraging users to utilize existing control features is scarce. It remains to be seen whether providing guidance and encouraging users to use these tools can actually change content outcomes. And if so, whether this active participation can lead to improved psychological outcomes such as satisfaction or perceived autonomy.

This study directly engages with these questions, building upon the foundation laid by prior research to empirically test the potentials and limits of user-driven algorithmic shaping.

Method

Participants

The study employed a one-group pretest-posttest design. All participants received the same intervention, and their outcomes were compared before and after the one-week period. Participants were 50 undergraduate students recruited from CUHK-Shenzhen. All the participants are active daily social media users.

Data Collection

The study lasted for one week. In the initial pre-test phase, all participants were asked to complete an online questionnaire. This questionnaire was designed to collect baseline data on users' levels of social media satisfaction, perceived control over their feed, frequency of social media use, and demographic information.

Following the pre-test, the participants attended a tutorial that explained the purpose and location of proactive control features on the platforms they used most. They were instructed to use these features consciously at least five times per day for a week, hiding content they found irrelevant, annoying, or undesirable.

After the one-week intervention phase, all participants were required to complete a post-test questionnaire that included some open-ended questions related to their experiences in the research.

Data Analysis

Quantitative data were analyzed using statistical tools. Paired-sample t-tests compared within-group changes, while independent-sample t-tests compared post-test scores between groups. Qualitative data were collected and analyzed using reflexive thematic analysis (Braun & Clarke, 2020) to identify recurring themes related to content changes and factors driving satisfaction.

Result

Quantitative Findings

A series of paired-samples *t*-tests were conducted to assess the effects of the one-week intervention requiring proactive algorithm control. The analysis compared participants' scores on key experience variables before and after the intervention. The results, presented in Table 1 below, indicate significant improvements across all measured dimensions.

Table1

Comparison of Pretest and Posttest Scores on Key Outcome Variables

Variable	Pre-test		Post-test		Paired-Samples *t*-test		
	M	SD	M	SD	*t* (49)	*p*	Cohen's *d*
Satisfaction	3.50	0.74	3.96	0.60	4.78	<.001	0.68
Content–Interest Match	3.40	1.05	4.11	0.81	5.27	<.001	0.75
Control Feeling	3.20	0.82	3.80	0.96	4.72	<.001	0.67

Note. N = 50. All variables were measured on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree).

As shown in Table 1, participant satisfaction with their social media experience increased significantly from pre-test to post-test. t (49) = 4.78, p < .001. The magnitude of this change was large. The most substantial improvement was in the content–interest match, which showed a very large increase from M = 3.40 to M = 4.11, t (49) = 5.27, p < .001. Participants also reported a significant increase in control feeling from M = 3.20 to M = 3.80.

Table 2

Post-Intervention Use and Perceptions of Proactive Control Features

Item	Category	Percentage (%)
Frequency of Control Feature Use	Sometimes	10
	Often	70
	Very Often	20
Ease of Using Control Features	Difficult	8

	Neutral	20
	Easy	58
	Very Easy	14
Perceived Degree of Content Change	Not at all	0
	A little	10
	Moderately	10
	Quite a bit	62
	A great deal	18

Note. N = 50. Data are from posttest questionnaire items regarding frequency, ease of use, and perceived impact on content.

The data in Table 2 shows that 90% of participants used the proactive control features during the week. A majority of participants found these features "Easy" or "Very Easy" to use. 80% of participants subjectively reported that using the features changed their content "Quite a bit" or "A great deal".

Qualitative Insights

The open-ended questions show three main themes. The first one is about the increased content relevance. Users reported that after dismissing unwanted content, their feeds became more relevant to their interests. The second one is enhanced autonomy. Many participants reported that feeling more in control made them feel less frustrated. The last one is about the reduced cognitive load. Many participants noted that seeing fewer irrelevant posts made their experience more pleasant and efficient.

Discussion

This study aimed to investigate whether and how active algorithm control influences social media satisfaction. The results provide clear answers to both research questions. First, they support the hypothesis that proactive algorithm control leads to measurable changes in content recommendations. Second, the increase in satisfaction appears to be driven by

enhanced perceived autonomy and improved content relevance. These findings align with prior research on algorithmic responsiveness to user feedback.

For user satisfaction, the increase appears to be driven by multiple factors. Primarily, the greater perceived control aligns with the core need for autonomy in Self-Determination Theory (Ryan & Deci, 2000), which is a key driver of psychological well-being and satisfaction. Improved content relevance further reinforced this positive experience.

However, the study has limitations. Regarding the research design, the sample was small and limited to university students, which may affect generalizability. Additionally, the study used a one-group design without a control group. This limits the ability to attribute changes solely to the intervention. Regarding the duration, the one-week intervention may be unable to capture long-term effects or potential adaptation by the algorithm. Furthermore, regarding data measurement, the self-reported data lack validity and are subject to bias.

Conclusion

This study demonstrates that proactive algorithm control can significantly alter content personalization and increase user satisfaction among social media users. By shifting attention from what algorithms do to users to what users can do to algorithms, it contributes to a more balanced, interactive understanding of human-algorithm relationships. The findings highlight the importance of autonomy and relevance for digital well-being, while also raising concerns about “filter bubbles” and the fragmentation caused by increased control. Future research should explore sustained intervention effects across varied populations and platform ecosystems. Practically, the results encourage social media designers to prioritize transparent, accessible control interfaces, which act as essential tools for fostering user agency, satisfaction, and active engagement in increasingly algorithmically mediated environments. In doing so, platforms may better align technical systems with human values, supporting more intentional and fulfilling digital experiences.

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Appendices

Pre-test questionnaire:

1. What's your name?
    - [Open-ended text field]
2. On average, how much time do you spend on your social media platform(s) per day?
    - [Single choice]
    - Options: 0~0.5h, 0.5h~1h, 1~2h, 2~3h, > 3h
3. Which social media platform(s) do you use most frequently?
    - [Multiple choice]
    - Options: Bilibili, 小红书 The Rednote, 抖音 TikTok, 快手, 微博 Weibo, X, YouTube, IG, Others [Details]
4. I understand how the platform's "For You" or recommendation algorithm works.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
5. I know how to use features (e.g., "Not interested", "Dislike") to influence my feed.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
6. I feel I have control over the content that is shown to me.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

7. I often use features to actively curate or control my algorithmic feed.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
8. The content on my feed accurately matches my interests.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
9. This is a check question. Please choose Strongly Agree.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
10. My experience on this platform is generally satisfying.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
11. The content I see makes me feel positive.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
12. After using this platform, I usually feel my time was well spent.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
13. In your own words, what is the one thing you wish you could change about the content recommended to you on your social media platform(s)?
    - [Open-ended text field, Optional]
14. Would you like to participate in a follow-up survey?
    - [Single choice]
    - Options: Yes, No

Post-test questionnaire

1. What's your name?
    - [Open-ended text field]
2. Did you participate in the research actively as required?
    - [Single choice]
    - Options: Yes, No
3. During the past week, how often did you use proactive control features (e.g., "Not interested", "Dislike")?
    - [Single choice]
    - Options: Never, Rarely, Sometimes, Often, Very often

If the participant chooses “No” in question 2, or “Never” “Rarely” and “Sometimes” in question 3, jump to question 16 directly.

4. How easy or difficult was it to use these control features?
    - [Single choice]
    - Options: Very difficult, Difficult, Neutral, Easy, Very easy
5. To what extent did you feel that using these features changed the content you saw?
    - [Single choice]
    - Options: Not at all, A little, Moderately, Quite a bit, A great deal
6. Which social media platform(s) do you use most frequently this week?
    - [Multiple choice]
    - Options: Bilibili, The Rednote, TikTok, 快手, Weibo, X, YouTube, IG, Others [Details]
7. My experience on this platform is generally satisfying.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

8. The content I see makes me feel positive.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
9. I often find the content on my feed to be meaningful.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
10. The content on my feed accurately matches my interests.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
11. This is a check question. Please choose Strongly Agree.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
12. After using this platform, I usually feel my time was well spent.
    - [Scale]
    - Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree
13. How, if at all, did using proactive control features (like “Not Interested”) change your social media experience over the past week?
    - [Open-ended text field]
14. Did you notice any changes in the type of content recommended to you? Please describe.
    - [Open-ended text field]
15. Is there anything else you would like to share about your experience with controlling your social media feed?
    - [Open-ended text field]
16. [Final single-choice question, context implied as continuation preference]

- [Single choice]
- Options: One week more